

### Quiz (Habanero)

- Q1.** A temperature increase of 3 degrees per day for 5 days results in a total temperature change of:  
(A)  $-15$  degrees  
(B) 0 degrees  
(C) 15 degrees  
(D) 25 degrees  
(E) 30 degrees
- Q2.** If it rains 2 inches every 4 days, how many inches will it rain in 12 days?  
(A) 1 inch  
(B) 2 inches  
(C) 3 inches  
(D) 4 inches  
(E) 6 inches
- Q3.** The pattern 1, 4, 7, 10, ... represents the number of species of plants in an ecosystem every 3 years. What is the next number in the pattern?  
(A) 10  
(B) 13  
(C) 16  
(D) 19  
(E) 22
- Q4.** If  $x = -5$ , then  $x \cdot (-2)$  is equal to:  
(A)  $-10$   
(B) 10  
(C) 15  
(D) 20  
(E) 25
- Q5.** A city's population is decreasing at a rate of  $-200$  people per year. If the current population is 10,000, what will the population be in 3 years?  
(A) 9,400  
(B) 9,600  
(C) 9,800  
(D) 10,200  
(E) 10,400
- Q6.** What is the value of  $x$  in the equation  $x + 2 = 7$ ?  
(A) 3  
(B) 5  
(C) 7  
(D) 9  
(E) 11
- Q7.** Which of the following numbers is an example of an odd integer?  
(A)  $-2$   
(B) 0  
(C) 2  
(D)  $-5$   
(E) 10
- Q8.** The number of inches of rainfall in a month is given by the formula  $R = 2n + 1$ , where  $n$  is the number of days. What is the rainfall in a month with 20 days?  
(A) 20 inches  
(B) 41 inches  
(C) 42 inches  
(D) 61 inches  
(E) 81 inches

#### Open Ended Questions (FRQ)

- Q9.** A glacier is melting at a rate of  $-0.5$  meters per year. If the current depth of the glacier is 100 meters, how many years will it take for the glacier to melt completely? Show your work.
- Q10.** The population of a city is increasing according to the pattern 1000, 1200, 1440, 1728, .... What is the next number in the pattern? Explain your reasoning and show your work.

#### Chef's Tips

- Chef's Tips
- Read each question carefully before answering
- Use a calculator only when necessary